Unit - 2

Surveying

* Surveying is the science and art of determining the relative positions of various points above, below or on the surface of earth
* The relative positions are determined by measuring horizontal distances, vertical distances, horizontal angles, vertical angles accurately using various surveying instruments.

Objects of surveying :-

* The primary objective of surveying is preparation of plan or map
* To take measurements to determine the relative positions of the existing features on or near the ground
* To layout or to mark the positions of the proposed structure on the ground
* To determine areas, volumes and other related quantities
* To determine dimensions and layouts of any part of the earth surface
* To establish boundaries b/w the land
* To accurate the calculation of areas for proposed structure
* To select a suitable sight for an engineering projects.
* To prepare a topographical map showing a details of hills, rivers and valleys
* To prepare a map of country showing the detailed location of cities, towns, villages and major roads.

Linear Measurements :-

* According to the standards of weights and measure ACT 1956, the units of measurement of length is in m, & cm

* prior to the introduction of metric units in India feet, tenth's & Hundrend's of a feet were used

Basic units of length :-

British units	Metric units
12 inches = 1 feet	10mm = 1 cm
3 feets = 1 yard	10cm = 1 decimeters
8 furlongs = 1 mile	10 decimeters = 1 meter
	10 meters = 1 decameter
	10 decameter = 1 hectometer
	10 hectometer = 1 kilometer

Conversion functions for length

Meters	Yards	Feet	Inches
1	1.0936	3.208	39.3701
0.9144	1	3	36
0.3048	0.333	1	12
0.0254	0.0278	0.0833	1

Basic units of Area :-

The units of measurements of area are sq. m, sq. cm & sq. mm

Basic units of Area

British units	Metric units
144 inch^2 = 1 ft^2	100 mm^2 = 1 cm^2
9 ft^2 = 1 Yard^2	100 cm^2 = 1 decimeter^2
1 Acre = 4840 yard	100 deci meter = 1 meter^2

100 meter^2 = 1 decameter^2

100 decameter^2 = 1 hectometer^2

100 hectometer^2 = 1 kilometer^2

Conversions functions for Area

Meters^2	Yards^2	Feet^2	Inches^2
1	1.196	10.7639	1550
0.8361	1	9	1296
0.0929	0.1111	1	144
0.00065	0.00077	0.0069	1

Basic units of volume :-

1. The units of measurement of volume in metric system are m^3, cm^3 & mm^3

British units

1728 inches^3 = 1 ft^3

27 ft^3 = 1 yard^3

Metric units

1000 mm^3 = 1 cm^3

1000 cm^3 = 1 decimeter^3

1000 decimeter^3 = 1 m^3

Angular measurements :-

1. An angle is the difference b/w in direction of two intersecting lines
2. Radian is the unit of plane angle
3. The radian is the angle b/w to radii of a circle which cut off the circumference
4. The circumference of an arc equal in length to radiu
5. There are three popular systems of angular measurement

Sexagesional system	centisimal system	Hours & system
1 circumference = 360° degree	1 circumference = 400 grades	1 circumference = 24hrs
1 degree = 60'	1grade = 100 centigrades	1hr = 60'
1 minute = 60''	1centigrade = 100 centi centi grade	1min = 60''

6. The sexagesional system are widely used in USA, Britain & India other parts of the world
7. The centisimal system is more favour in Europe
8. The hours system is mostly used in navigation
9. The basic units measurements of an angle is radians

Conversions of Radians to Degrees :-

π radians = 180° degrees

1 radian = $\frac{180°}{\pi}$ degree

1 Degree = $\frac{\pi}{180}$ radians

$\pi = \frac{22}{7}$ or 3.14

1 radian = $\frac{180}{3.14}$ deg

= 57.324°

* 1 person move from point A towards North by the distance 10mts and he travelled towards east by 20mts Determine the shortest distance from two terminal points. Assume the angle b/w the lines is 60°

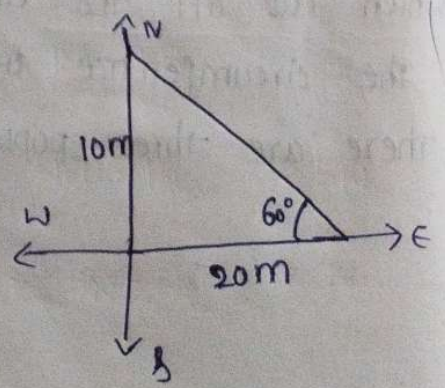
shortest distance =

$\tan\theta = \frac{x}{10}$

$\tan 60 = \frac{x}{10}$

$\tan 60 = 10\sqrt{3}$

$= 10\sqrt{3}$ m $= 17.32$m



Bearings :-

The bearing of a line is the horizontal angle which makes w.r.to reference line (meridian) depends upon meridian

Types of bearings :-

1. True bearing
2. Magnetic bearing
3. Grid bearing
4. arbitary bearing

1. True bearing :-

* The true bearing of a line is the horizontal angle b/w the meridian (or) true survey line
* The true bearing measured from the north in the clockwise direction

2. Magnetic bearing :-

* The magnetic bearing of a line is the horizontal angle which the line makes with the magnetic north

Grid bearing :-

* The Grid bearing of a line is the horizontal angle which the line makes with grid meridian.

Arbitary bearing :-

* The arbitary bearing of a line is the horizontal angle which the line makes with arbitary meridian

Desgination of bearings :-

The bearings are desginated in following two system

1. Whole circle bearing system (W.C.B)
2. Quadrantal bearing system Q.B

Whole circle bearing system :-

The bearing of line measured w.r.to North of magnetic meridian in clockwise direction is called W.C.B or Magnetic bearing

The values varies b/w 0° to 360°

* The quadrants start from N and progress in a clockwise direction as the first quadrant is 0° to 90° in clockwise direction (c.d) and second quadrant b/w 90° to 180° third quadrant b/w 180° to 270° & fourth quadrant b/w 270° to 360° in C.D

Quadrantal Bearing system (Q.B) :-

* In this system the bearings of survey lines are measured w.r.to North line or south line which ever is the nearest to the given Survey line and either in clockwise direction or Anti clockwise direction
* The bearing lines which fall I and IV quadrants are measured w.r.to north line because north line is the nearer than South line and bearing of lines in fall in II and III quadrants are measured from south line. As South line is nearer than north line

Reduced bearing :- (R.B)

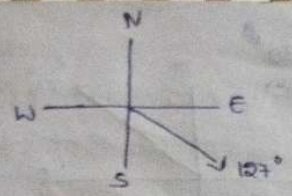
* when the W.C.B of a line is converted into Q.B it is termed as reduced bearing (R.B)
* The R.B is similar to Q.B its value is lies b/w 0° to 90° but Quadrants Should be mentioned for proper designation

Conversion of WCB to Q.B

case	Range of WCB	conversion	Quadrants
1.	0° to 90°	RB = WCB	NE
2.	90° to 180°	RB = 180° - WCB	SE
3.	180° to 270°	RB = WCB - 180°	SW
4.	270° to 360°	RB = 360° - WCB	NW

Ex:-

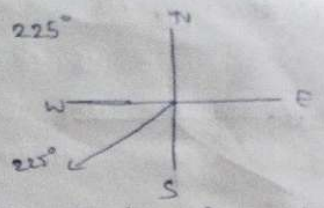
(i) 127°



RB = 180° - 127° = 53°

Q.B = S 53° E

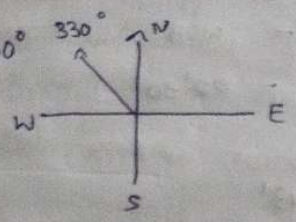
(ii) 225°



R.B = 225° - 180° = 45

Q.B = S 45° W

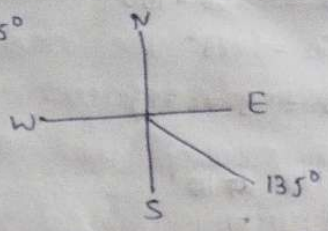
(iii) 330°



R.B = 360° - 330° = 30°

Q.B = N 30 W

(iv) 135°



R.B = 180° - 135° = 45°

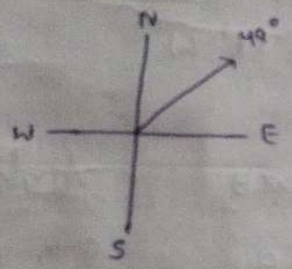
Q.B = S 45° E

* convert the following W.C.B into R.B

(i) 49° (ii) 240 (iii) 133 (iv) 335 (v) 22° 30' (vi) 135°

(vii) 345 (viii) 149 (ix) 256

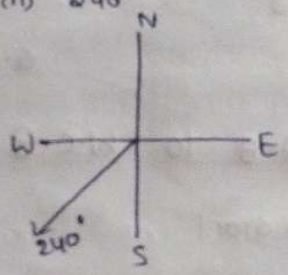
Sol (i) 49°



R.B = 49°

Q.B = N 49° E

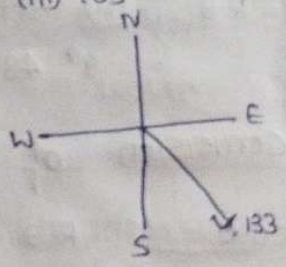
(ii) 240°



R.B = 240° - 180°
= 60°

Q.B = S 60° W

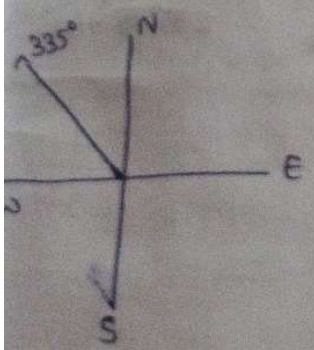
(iii) 133



R.B = 180 - 133
= 47°

Q.B = S 47° E

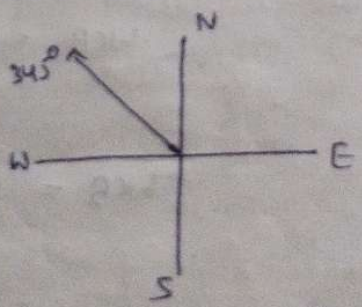
(iv) 335



R.B = 360° - 335°
= 25°

Q.B = N 85° W

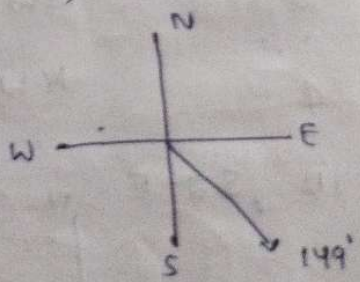
(vi) 345°



R.B = 360° - 345°
= 15°

Q.B = N 15° W

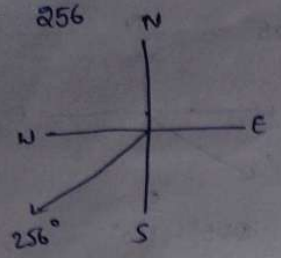
7) 149



R.B = 180 - 149
= 31°

Q.B = S 31 E

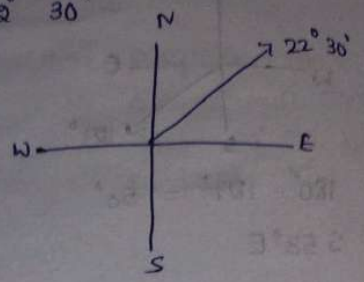
8. 256



R.B = 256 - 180
= 76°

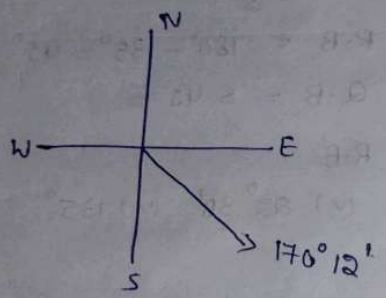
Q.B = S 76° W

9. 22° 30'



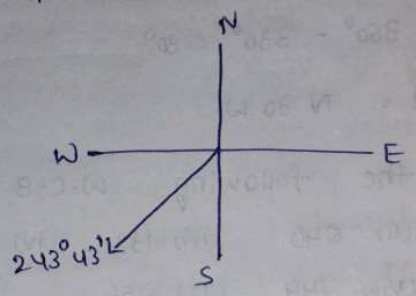
R.B = WCB
= 22° 30'

* 170° 12'



R.B = 180 - 170° 12'
= 9° 48'

* 243° 43'

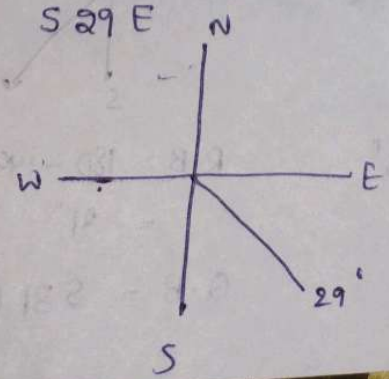


R.B = 243° 43' - 180°
= 62° 17'
Q.B = S 62° 17' W

Conversion of Q.B to W.C.B

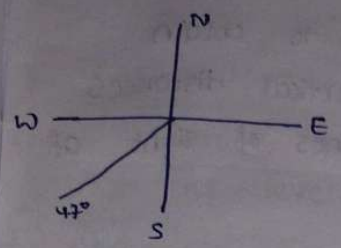
case	R.B quadrant	Rule for WCB	Angles
1	NE	WCB = Q.B	0 to 90°
2	SE	WCB = 180 - Q.B	90 to 180°
3	SW	WCB = 180 + QB	180° to 270°
4	NW	WCB = 360 - QB	270° to 360°

(i) S 29 E



WCB = 180 - QB
= 180 - 29
= 151°

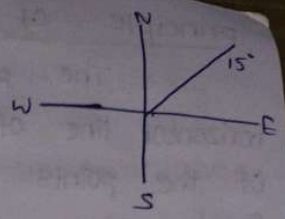
(ii) S 47° W



WCB = 180 + Q.B
= 180 + 47
= 227°

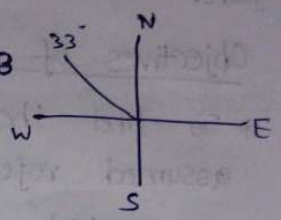
(iii) N 15° E

WCB = Q.B
WCB = 15°

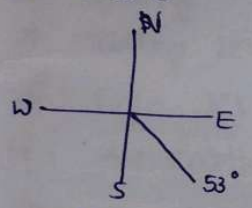


(iv) N 33° W

WCB = 360 - QB
= 327°

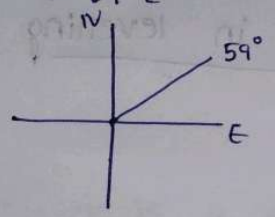


(v) S 53° E



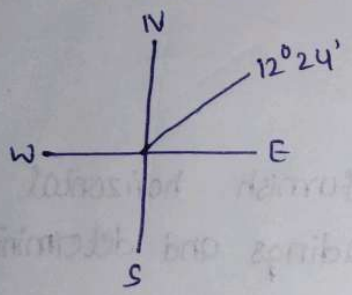
WCB = 180 - QB
= 180 - 53
= 127°

(vi) N 59° E



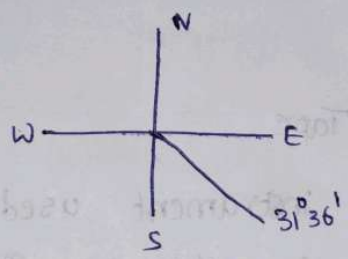
WCB = Q.B
= 59°

* N 12°24' E



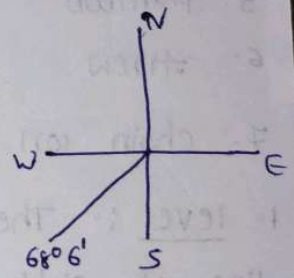
WCB = Q.B
= 12° 24'

* S 31°36' E



WCB = 180 - Q.B
= 180 - 31°36'
= 148° 24'.

* S 68°6' W



WCB = 180 + Q.B
= 180 + 68°6'
= 248° 6'

Levelling :-

Levelling is defined as an art of determining the relative tights of different points on above or below the surface.

principle of levelling :-

The principle of levelling is to obtain horizontal line of sight w.r.to which vertical distances of the points above (or) below these lines of sight found

Objectives of levelling :-

* To find the elevation of different points w.r.to assumed reference line called Datum
* To establish point at required elivation w.r.to datum

Instruments used in levelling :-

1. level
2. levelling staff
3. Tripod
4. Rangeing Rod
5. plumbob
6. Arrow
7. chain (or) Tape

1. **level** :- The instrument used to furnish horizontal line of sight for observing staff readings and determining reduced levels (RLS)

2. **levelling staff** :- levelling staff or scales and which vertical distances are measured.

3. **Tripod** :- Tripod is a three leged stand designed to support level

4. **Rangeing Rod** :- It is a surveying instrument used for marking the positions of stations and for sighting of those stations as well as for rangeing straight lines

5. plumbob :-

It is used to fix the level in correct way

6. Arrow :-

Arrows are the metal spikes used to mark the position of level.

7. Chain (or) Tape :-

It is used to determine horizontal distances

Types of levels :-

1. Dumpy level
2. Tilting level
3. Wye level
4. Automatic level

Dumpy level :- Dumpy level is a simple, compact & stable instrument. The telescope is regidly fixed to support. Hence it cannot be rotated above horizontal ax

Tilting level :- In this level the telescope fills about horizontal axis hence it is called tilting level.

S.No	B.S	I.S	F.S	H.I	R.L	Remarks
1.	1.60			1051.6	1050	HI = RL+BS
2.		1.90	(−)		1049.7	RL = HI − IS
3.		1.05			1050.5	"
4.		2.0			1049.6	
5.		2.35			1049.2	
6.		1.70			1049.2	
7.		1.90			1049.9	
8.		2.80			1048.8	
9.		3.20			1048.4	
10.		3.80			1047.0	
11.		0.90	~~0.60~~		1050.7	
			0.60		1051	RL = HI − FS

check :-

$\Sigma BS - \Sigma FS$ = Last RL − first RL

$1.60 - 0.6 = 1051 - 1050$

$1 = 1$

2. The following readings for taken from a dumpy level 2.0, 2.65, 3.20, 1.80, 1.70, 1.60, 2.70, 3.20, 0.80, 1.15, 0.50 the RL of bench mark of 50 m do levelling with height of instrument method.

S.NO	B.S	I.S	F.S	H.I	RL	Remarks
1.	2.0			52	50	HI = RL+BS
2.		2.65			49.35	RL = HI - IS
3.		3.20			48.8	"
4.		1.80			50.2	
5.		1.70			50.3	
6.		1.60			50.4	
7.		2.70			49.3	
8.		3.20			48.8	
9.		0.80			51.2	
10.		1.15			50.85	
11.			0.50		51.5	RL = HI - FS

$\Sigma BS - \Sigma FS$ = Last RL - first RL

$2.0 - 0.50 = 51.5 - 50$

$1.5 = 1.5$

3. The following readings for taken from a dumpy level 2.0, 2.65, 1.60, 1.90, 1.05, 2.0, 2.35, 1.70, 1.90, 2.80, 3.20, 3.80, 0.90, 0.60 the RL of bench mark 1050 meters do levelling with raise & fall method.

S.NO	B.S	I.S	F.S	Rise	Fall	RL	Remarks
1.	1.60					1050	
2.		4.90			0.3	1049.7	
3.		1.05		0.85		1050.55	
4.		2.0			0.95	1049.6	
5.		2.35			0.35	1049.25	
6.		1.70		0.65		1049.9	
7.		1.90			0.20	1049.7	
8.		2.80			0.9	1048.8	
9.		3.20			0.4	1048.4	
10.		3.80			0.6	1047.8	
11.		0.90		2.9		1050.7	
12			0.60	0.3		1051	

EBS − EFS = Σ Rise − Σ fall = last RL − first RL

1.60 − 0.60 = 4.7 − 3.7 = 1050 − 1050

1 = 1 = 1

4. The following readings for taken from a dumpy lev
2.0, 2.65, 3.20, 1.80, 1.70, 1.60, 2.70, 3.20, 0.80
1.15, 0.50 the RL of bench mark is 50 do
leveling with raise & fall method.

S.NO	B.S	I.S	F.S	Raise	Fall	R.L	Remarks
1.	2.0					50	
2.		2.65			0.65	49.35	
3.		3.20			0.55	48.8	
4.		1.80		1.4		50.2	
5.		1.70		0.1		50.3	
6.		1.60		0.1		50.4	
7.		2.70			1.1	49.3	
8.		3.20			0.5	48.8	
9.		0.80		2.4		51.2	
10.		1.15			0.35	50.9	
11.			0.50	0.65		51.55	

$\Sigma BS - \Sigma FS = \Sigma Rise - \Sigma Fall =$ Last RL - First RL

$2.0 - 0.50 = 4.15 - 3.1 = 51.55 - 50$

$1.5 = 1.5 = 1.5$

5. The following readings were taken with dumpy level 0.894, 1.643, 2.596, 3.016, 0.954, 0.692, 0.582, 0.251, 1.532, 0.996, 2.135 the instrument was shifted after 4th & 8th reading the 1st reading was taken on the staff held on the bench mark of RL 820.765m calculate the reduced level of points by using raise and fall method and show the usual check.

B.S	I.S	F.S	Rise	Fall	RL	Remarks
0.894					820.765	BS - IS
	1.643			0.749	820.016	or
	2.896			1.253	818.763	IS - IS
0.954		3.016		0.120	818.643	or
	0.692		0.262		818.905	IS - FS
	0.582		0.110		819.015	
1.532		0.251	0.331		819.346	
	0.996		0.536		819.882	
		2.135		1.139	818.743	

$\Sigma BS - \Sigma FS = \Sigma Rise - \Sigma Fall = Last\ RL - First\ RL$

$3.380 - 5.402 = 1.239 - 3.261 = 818.743 - 820.765$

$-2.022 = -2.022 = -2.022$

Height of instrumentation method

B.S	I.S	F.S	H.I	RL	Remarks
0.894			821.659	820.765	HI = RL + BS
	1.643			820.016	
	2.896			818.763	RL = HI - IS
0.954		3.016		818.643	
	0.692		819.597	818.905	RL = HI - FS
	0.582			819.015	
1.532		0.251	820.878	819.346	
	0.996			819.882	
		2.135		818.743	

$\Sigma BS - \Sigma FS = Last\ RL - first\ RL$

$=$